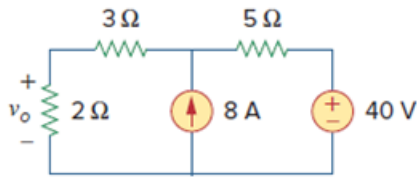


Problem

Using the superposition theorem, find v_o in the circuit of Fig. 4.8.

Figure. 4.8:

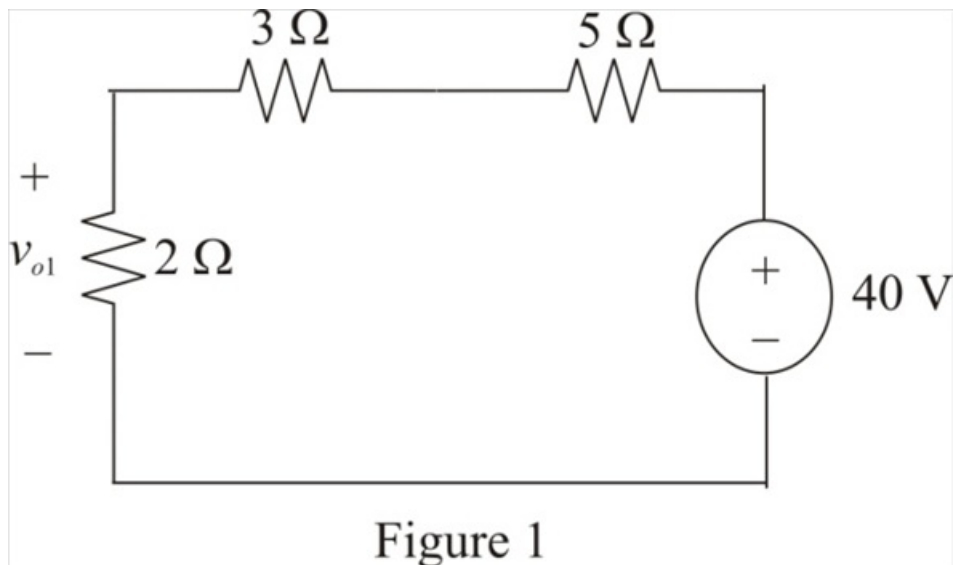


Step-by-step solution

Step 1 of 5

Refer to Figure 4.8 in the text book.

First deactivate the current source from the circuit diagram as shown in Figure 1.



Step 2 of 5

Calculate the value of the voltage, v_{o1} .

$$\begin{aligned} v_{o1} &= (40) \left(\frac{2}{2+3+5} \right) \\ &= (40) \left(\frac{2}{10} \right) \\ &= (40)(0.2) \\ &= 8 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_{o1} is 8 V.

Step 3 of 5

Deactivate the voltage source from the circuit diagram as shown in Figure 2.

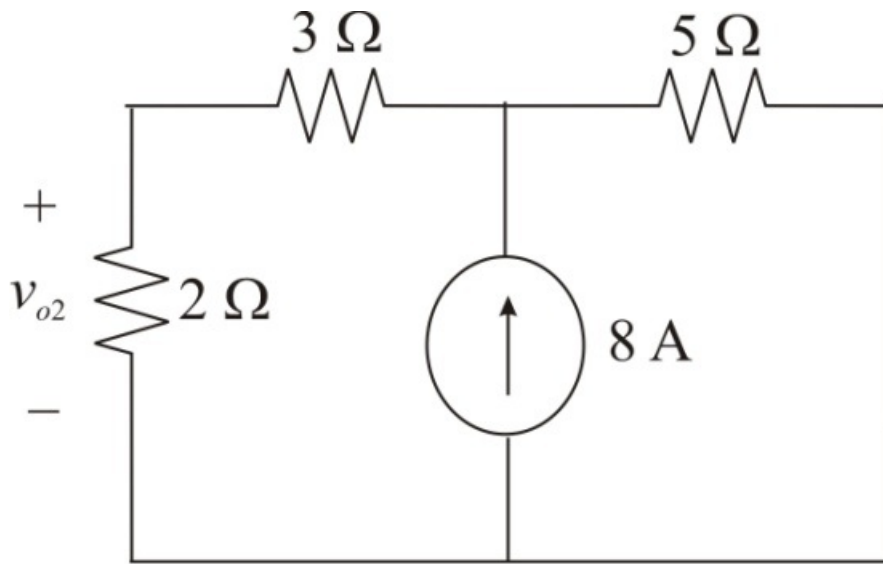


Figure 2

Step 4 of 5

Calculate the current in $2\ \Omega$ resistor.

$$\begin{aligned} i_{2\ \Omega} &= (8) \left(\frac{5}{2+3+5} \right) \\ &= (8) \left(\frac{5}{10} \right) \\ &= (8)(0.5) \\ &= 4\ \text{A} \end{aligned}$$

Calculate the value of the voltage, v_{o2} .

$$\begin{aligned} v_{o2} &= (i_{2\ \Omega})(2) \\ &= (4)(2) \\ &= 8\ \text{V} \end{aligned}$$

Therefore, the value of the voltage v_{o2} is $8\ \text{V}$.

Step 5 of 5

Calculate the value of the voltage, v_o .

$$v_o = v_{o1} + v_{o2}$$

Substitute $8\ \text{V}$ for v_{o2} and $8\ \text{V}$ for v_{o1} in the equation.

$$\begin{aligned} v_o &= 8 + 8 \\ &= 16\ \text{V} \end{aligned}$$

Therefore, the value of the voltage v_o is 16 V.